

Technical Requirements Document (TRD)

No Food Waste — Food Rescue & Redistribution Platform

1. Recommended Technology Stack

- Frontend: React + Vite + Tailwind CSS
- Backend: Node.js + Express.js
- Database: MongoDB Atlas + Mongoose
- Authentication: JWT + bcrypt (Roles: DONOR, VOLUNTEER, ADMIN)
- Real-Time Communication: Socket.IO for request-scoped chat and notifications
- Maps & Location: MapLibre GL JS / Leaflet, Geocoding, reverse geocoding, route calculations
- File Storage: Object storage for delivery proofs and photos, URLs saved in MongoDB

2. Core API Specifications

Method	Endpoint	Access	Description
POST	/api/auth/register	Donor/Volunteer	Register account
POST	/api/auth/login	All	Login and receive JWT
GET	/api/auth/me	All	Current user & role profile
POST	/api/requests	Donor	Create donation request with location & food details
GET	/api/requests/mine	Donor	List own requests
GET	/api/requests/available	Volunteer	Available requests in region, sorted by priority score
POST	/api/requests/:id/accept	Volunteer	Atomic request claiming by volunteer
POST	/api/requests/:id/delivery-proof	Volunteer	Upload GPS location + food + delivery spot photos
GET	/api/admin/analytics/overview	Admin	Food totals, deliveries, and request metrics
GET	/api/admin/analytics/hotspots	Admin	Aggregated delivery coordinates for hunger heatmap

3. Priority Engine & Hotspot Intelligence

Priority Scoring Formula: $Priority\ Score = (Distance\ Weight \times Distance\ Score) + (Quantity\ Weight \times Quantity\ Score)$.
Weights are centrally configurable.

Hunger Hotspots: Delivery coordinates are aggregated using geospatial density clustering to identify recurring delivery concentrations and visualize high-need zones.